

Motor Voltage and Current Requirements and Corresponding Power Electronics

The selected motor is the Pololu 50:1 Micro Metal Gearmotor HP 6V. The detailed values of this motor are:

No load speed $n_{no\ load}$ [rpm]	590
No load current I_0 [A]	0.1
Stall current I_{stall} [A]	1.6
Stall torque T_{stall} [Nm]	0.084
Supply voltage V_s [V]	6

For a brushed DC motor, we know that:

$$V_s = IR + k_b n$$

Where,

I = motor current [A]

R = armature resistance [ohm]

k_b = back-EMF constant [V/(rad/s)]

n = motor rotational speed [rpm]

$$T = k_t I$$

Where,

T = motor torque [Nm]

k_t = torque constant [Nm/A]

Combining them shows that the torque speed curve is linear and implies that torque is proportional to current for a fixed motor and supply. At stall, $n = 0$.

$$I_{stall} = \frac{V_s}{R}$$

$$T_{stall} = k_t I_{stall}$$

So for a first-order estimate,

$$\frac{T}{T_{stall}} \approx \frac{I - I_0}{I_{stall} - I_0}$$

$$I \approx I_0 + \left(\frac{T}{T_{stall}} \right) (I_{stall} - I_0)$$

Then we can apply our worst-case requirement into it.

$$n_{req} = 400 \text{ rpm}, \quad T_{req} = 0.02058 \text{ Nm}$$

Then we can get the estimate operating current

$$I_{req} \approx I_0 + \left(\frac{T_{req}}{T_{stall}} \right) (I_{stall} - I_0) \approx 0.466 \text{ A}$$

This is well below the 1.6 A stall current, which is consistent with operating away from stall.

Then we do the power check. For the mechanical power at worst-case point,

$$P_{mech} = T\omega$$

Where,

P_{mech} = mechanical power output [w]

ω = angular velocity [rad/s]

So we can calculate:

$$\omega = \frac{2\pi n}{60} \approx 41.89 \text{ rad/s}$$

$$P_{mech} = 0.862 \text{ W}$$

Electrical input power estimate:

$$P_{elec} = V_s I_{req} = 2.80 \text{ W}$$

The efficiency is about:

$$\frac{P_{mech}}{P_{elec}} = \frac{0.862}{2.80} = 31$$

Which is reasonable for a small brushed DC gearmotor.

For the power electronics requirements and part selection. Because the motor can draw up to stall

current during setup, jam or aggressive acceleration, the driver must be sized for:

- Motor supply: 6V
- Continuous operating current $\approx 0.47\text{ A}$ (worst-case)
- Peak current capability $\geq 1.6\text{ A}$ (stall)

Recommended Driver

A suitable choice is an H bridge driver rated for $\geq 2\text{ A}$ peak, TB6612FNG-class driver is recommended.

This supports:

- PWM speed control
- Direction control
- Peak current headroom above 1.6 A

Power Supply Requirement

The motor supply must provide 6V and support both:

- Normal operation: $\sim 0.5\text{ A}$
- Transient peak: up to 1.6 A

This can be achieved using a 6V DC adapter rated $\geq 2\text{ A}$.